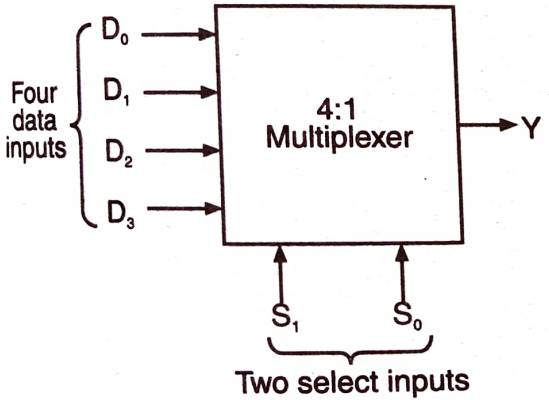


Fig. 6.8. 4 : 1 Multiplexer

Table 6.3

Select inputs		Output
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Here, it may be noted that n = 4, hence, number of select lines, i.e., m = 2, so that $2^m = n$. The strobe terminal (G) has not been included to reduce the complexity. The truth table tells us that if $S_1 S_0 = 00$, the data bit D_0 is selected and routed to output

Therefore, we have $Y = D_0$ when $S_1 S_0 = 00$

Similarly, if $S_1 S_0 = 01$, then D_1 is selected and routed to the output.

Therefore, $Y = D_1$ when $S_1 S_0 = 01$

Also, $Y = D_2$ for $S_1 S_0 = 10$ and $Y = D_3$ for $S_1 S_0 = 11$

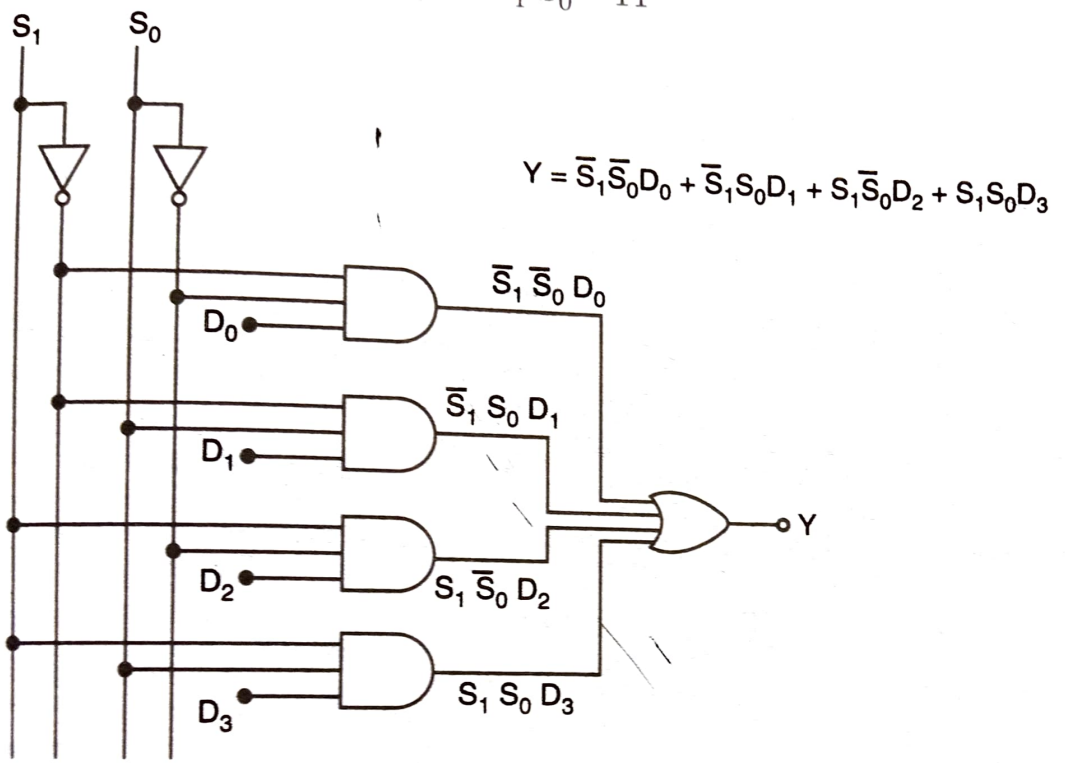


Fig. 6.9. Realization of 4 : 1 multiplexer using basic gates

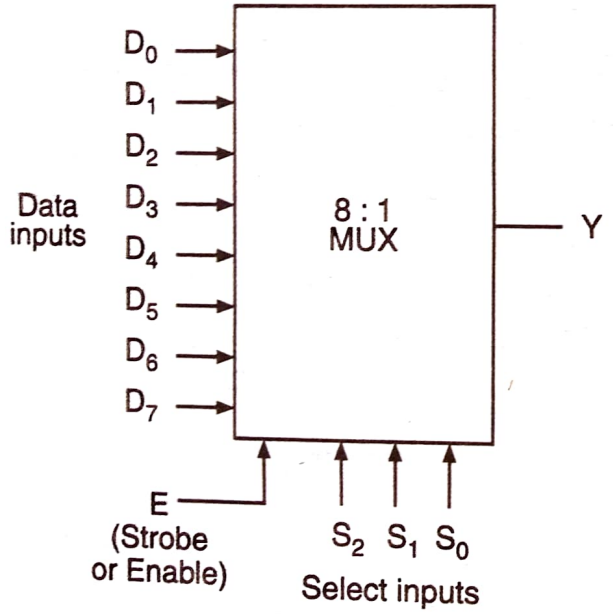
The output will be high when the selected input (D_0, D_1 etc.) is 1. Hence, the logical expression for output in the SOP form will be as under :

$$Y = \overline{S_1}\ \overline{S_0}\ D_0 + \overline{S_1}\ S_0\ D_1 + S_1\ \overline{S_0}\ D_2 + S_1\ S_0\ D_3$$

This expression can be realized using basic gates as shown in figure 6.64.

6.6.2. A 8 : 1 Multiplexer

The block diagram of an 8 : 1 MUX has been shown in figure 6.10 (a) and its truth table is shown in figure 6.10 (b). It has eight data inputs, one enable input, three select inputs and one output.



(a) Block diagram

Enable	Select inputs			Output
E	S_2	S_1	S_0	Y
0	×	×	×	0
1	0	0	0	D_0
1	0	0	1	D_1
1	0	1	0	D_2
1	0	1	1	D_3
1	1	0	0	D_4
1	1	0	1	D_5
1	1	1	0	D_6
1	1	1	1	D_7

(b) Truth table

Fig. 6.10. 8 : 1 Multiplexer

Operating principle

When the strobe or enable input is 0, the output will be 0 irrespective of any input. With E = 1 we can select any one of the eight data inputs and connect it to the output. For example, if S_2 S_1 S_0 = 0 1 1, then the data input D_3 is selected and output Y will follow the input D_3.

Realization using gates

Realization using gates

The realization of 8 : 1 MUX using gates has been shown in figure 6.11.

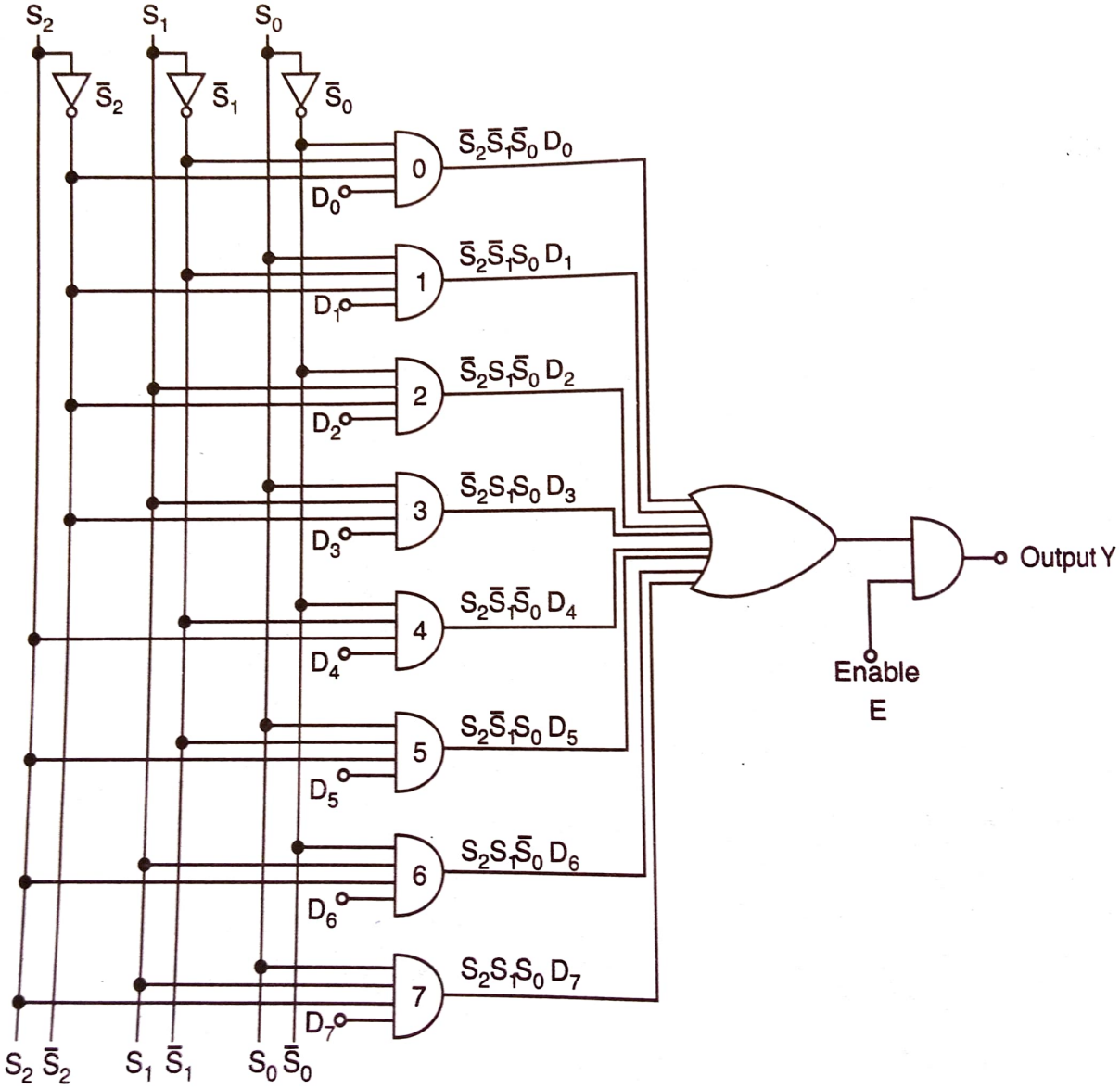


Fig. 6.11. Illustration of a 8 : 1 MUX using gates

6.7. APPLICATIONS OF A MULTIPLEXER

Some of the important applications of a multiplexer may be listed as under :

(i) It is used as a data selector to select one out of many data inputs.

(ii) It is used for simplification of logic design.

(iii) In the data acquisition system.

(iv) In designing the combinational circuits.

(v) In the D/A converters.

(vi) To minimize the number of connections.

EXAMPLE 6.3. Obtain an 8 : 1 multiplexer using two 4 : 1 multiplexers.

Solution: The cascading of two 4 : 1 multiplexer results in 8 : 1 multiplexer as shown in figure 6.13.

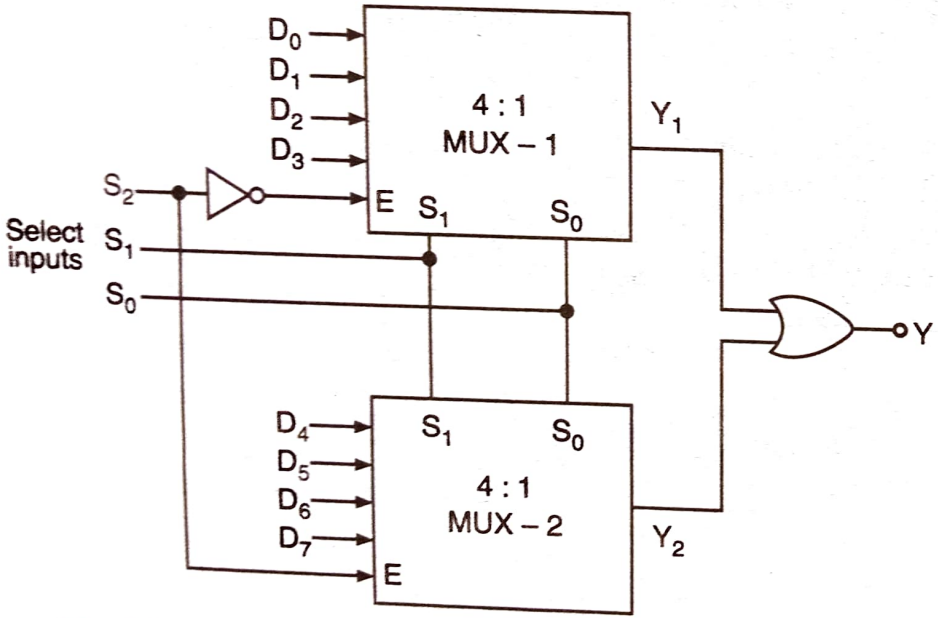


Fig. 6.13. 8 : 1 multiplexer by cascading two 4 : 1 multiplexers

Table 6.5. Truth table

Select inputs			Output	
S_2	S_1	S_0	Y	
0	0	0	D_0	MUX 1 is enable
0	0	1	D_1	
0	1	0	D_2	
0	1	1	D_3	
1	0	0	D_4	MUX 2 is enable
1	0	1	D_5	
1	1	0	D_6	
1	1	1	D_7	

There are in all eight data inputs (D_0 through D_7). The select lines S_1 and S_0 of both 4 : 1 multiplexers are connected in parallel whereas a third select input S_2 is used for enabling one multiplexer at a time. Also, S_2 is connected directly to the enable (E) terminal of MUX-1 whereas $\overline{S}_2$ is connected to the enable terminal of MUX-2. The outputs of the two multiplexers are ORed to obtain the final output Y.

EXAMPLE 6.4. Implement a 16 : 1 multipl

EXAMPLE 6.4. Implement a 16 : 1 multiplexer using 4 : 1 multiplexers.

Solution: Let us consider figure 6.14 for implementation.

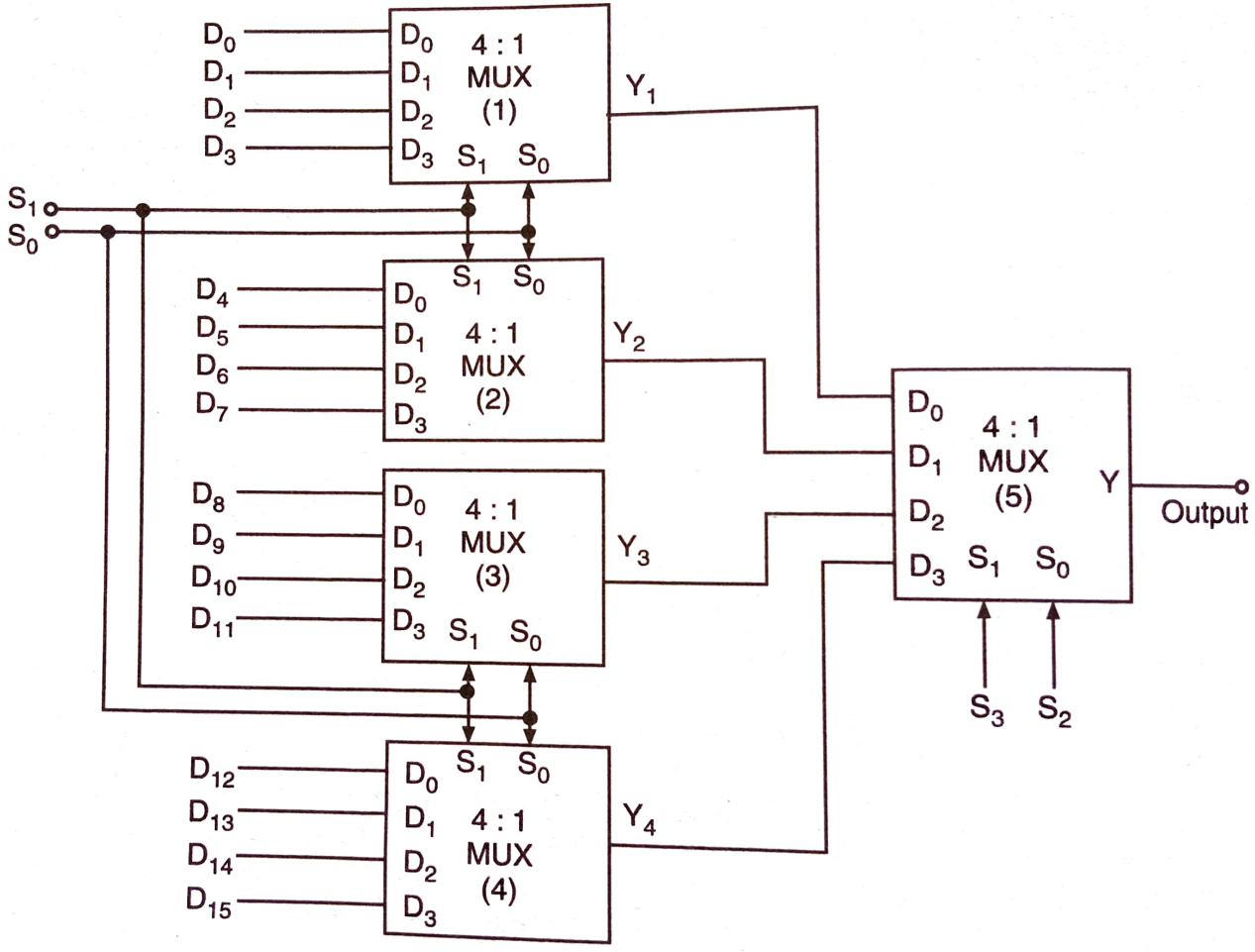


Fig. 6.14. 16 : 1 multiplexer using 4 : 1 multiplexers

6.10. DEMULTIPLEXER

The block diagram of a demultiplexer or decoder is shown in figure 6.38 (a). It has only one input, n outputs, and m select inputs. A demultiplexer performs the reverse operation of a multiplexer, i.e., it receives one input and distributes it over several outputs. At a time, only one output line is selected by the select lines and the input is transmitted to the select output line.

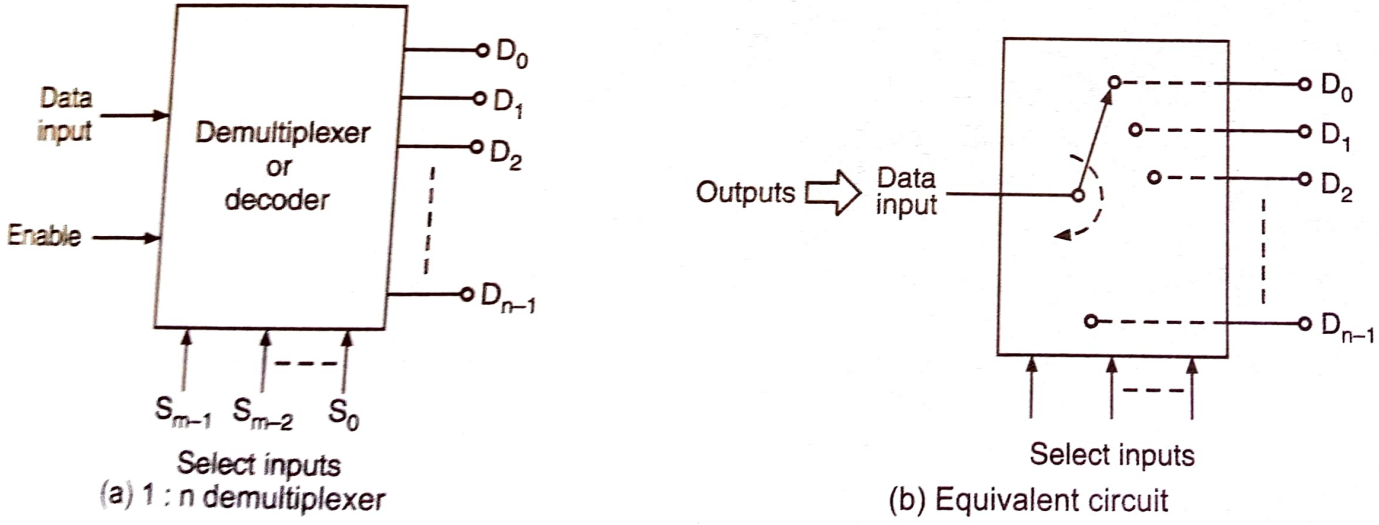


Fig. 6.38.

Hence, a demultiplexer is equivalent to a single pole multiple way switch as shown in figure 6.38 (b). The enable input will enable the demultiplexer. The relation between the n output lines and select lines is as under :

$$n = 2^m.$$

6.11. CLASS

Realization of 1 : 2 Demux

(i) To realize the 1 : 2 demultiplexer using gates, we have to write the truth table as under :

Table 6.14. Truth table of demux 1 : 2

Enable	Data	Select	Outputs	
E	D_{in}	S_0	Y_1	Y_0
0	×	×	0	0
1	0	0	0	0
1	1	0	0	1
1	0	1	0	0
1	1	1	1	0

$\longrightarrow$ Y_0 is connected to D_{in}

$\longrightarrow$ Y_1 is connected to D_{in}

(ii) Next, we write the K-maps.

The maps for the two outputs are as under :

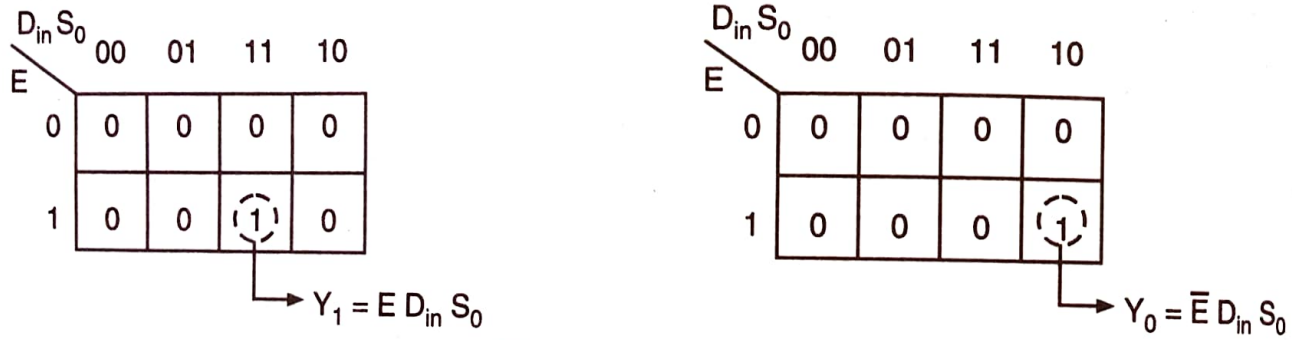


Fig. 6.39. K-maps

Thus, the expressions for Y_0 and Y_1 are as under :

$$Y_0 = ED_{in}\,\bar{S}_0 \quad \text{and} \quad Y_1 = E\,D_{in}\,S_0$$

The 1 : 2 demux using gates has been shown in figure 6.40.

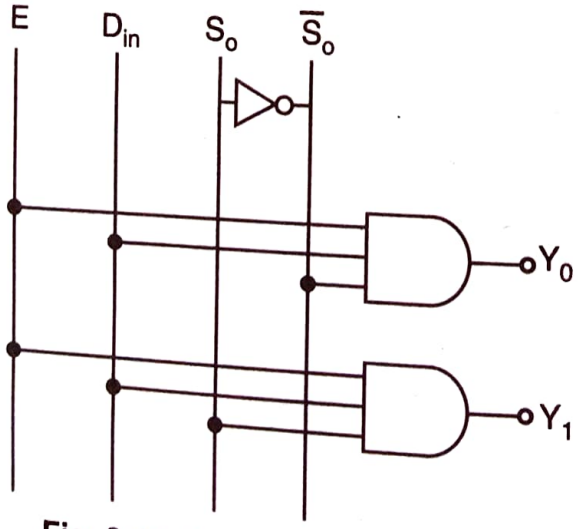


Fig. 6.40. 1 : 2 demux using gates

6.13. PERFORMANCE COMPARISON OF MULTIPLEXER AND DEMULTIPLEXER

Table 6.21. Performance Comparison of MUX and DEMUX

Sr. No.	Parameter of comparison	Multiplexer	Demultiplexer
1.	Type of logic circuit	Combinational	Combinational
2.	Number of data inputs	n	1
3.	Number of select inputs	m	m
4.	Number of data output	1	n
5.	Relation between input/output lines and select lines	$n = 2^m$	$n = 2^m$
6.	Operation principle	Many to 1 or as data selector	1 to many or data distributor
7.	Application	(i) As a universal logic circuit, we can implement any combinational circuit using a MUX. (ii) In time division multiplexing at the sending end.	(i) We can implement some combinational circuit (ii) In TDM system at receiving end.